Challenges in Dispersing and Compounding Carbon Nanotubes and Graphene into Thermoplastic Polymers

and Nanoclays

Brian P. Grady

Director, Institute for Applied Surfactant Research

Fellow, Society of Plastics Engineers

Conoco-DuPont Professor and Director

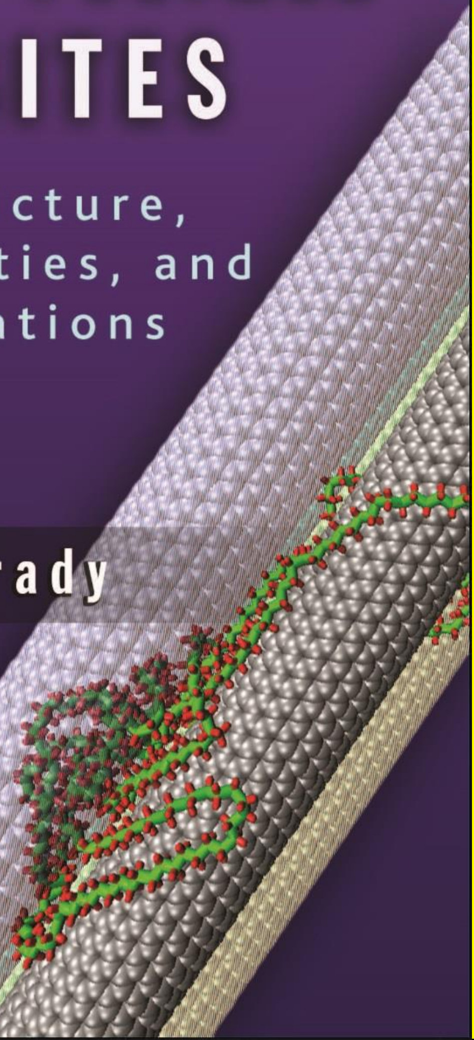
School of Chemical, Biological and Materials Engineering

University of Oklahoma

bpgrady@ou.edu

CARBON NANOTUBE-POLYMER COMPOSITES

Manufacture, Properties, and Applications



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Thermoplastic polymers

- Polyethylene (various forms: HDPE, LDPE,LLDPE …), polypropylene, polyethylene terephthalate (PET), nylon, polycarbonate, polystyrene (styrofoam), polymethyl methacrylate (PMMA), polyvinyl chloride (PVC)
- 300 million metric tons produced per year or about 90 pounds per person per year for every person on the planet!!!! Growing ~3% year (population growth ~1%/year)
- Focus on commercially-viable solutions for this market

Thermoplastic Polymers are a low-margin business so price is king!

Nanoclays, Nanotubes, Graphene

- Nanoclays
  - Clays are very commonly used in thermoplastic polymers to reduce cost (cheaper than dirt!)
  - Nanoclays are a very specialized subset of clays
    - Montmorillonite (Cloisite®)
    - Need cationics surfactant for dispersion ($5/pound)
- Nanotubes
  - Single-walled or multi-walled
  - Only multi-walled used in thermoplastic polymers (worldwide 1 million pounds/year, growing at 20%/year)
- Graphene
  - Pristine graphene or graphene oxide
  - Volume in thermoplastic polymers ~0

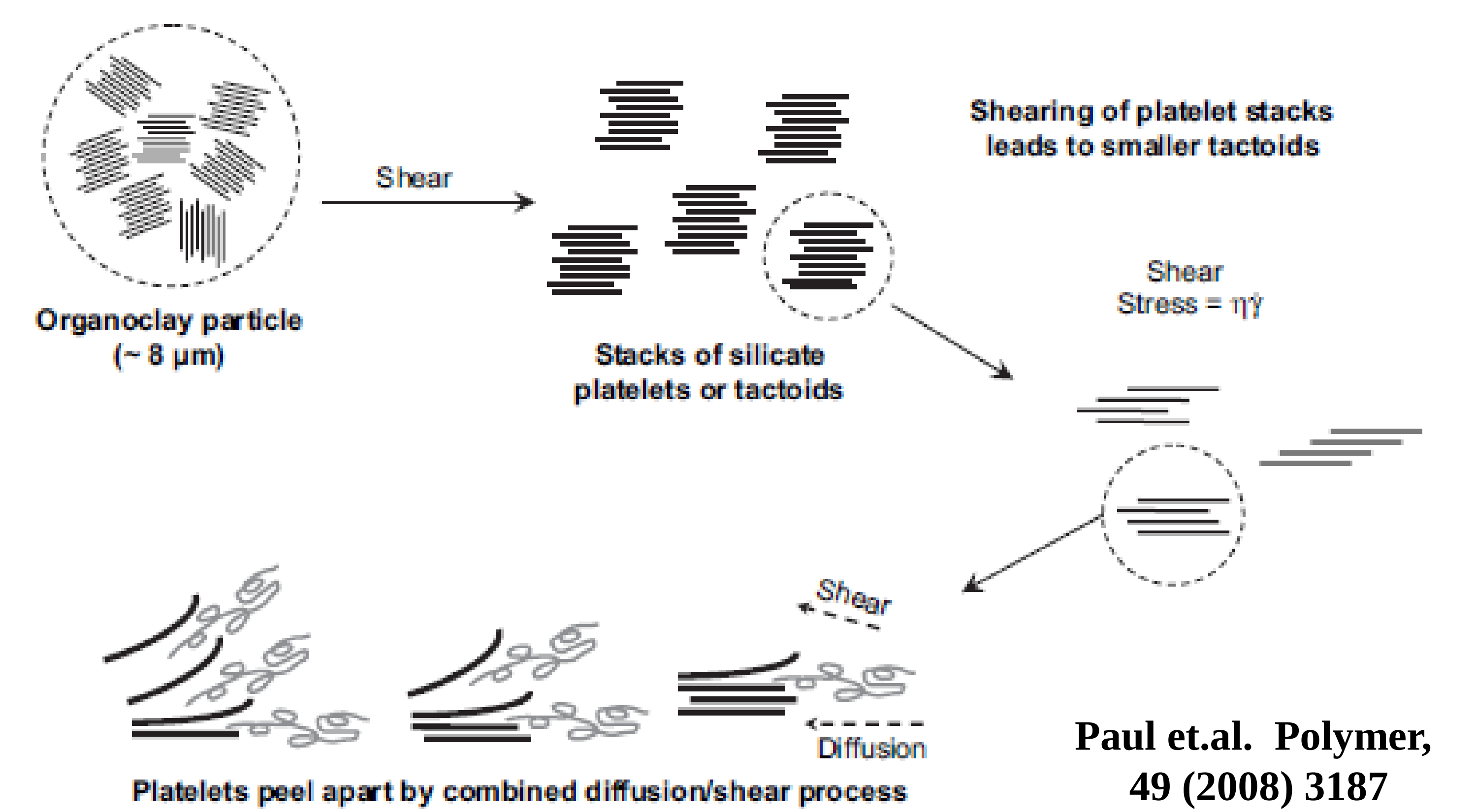
Why put Nanoclays, Nanotubes, Graphene in Thermoplastic Polymers

- Increase stiffness: Not been very successful because of cost and performance at higher loadings (competition: glass fibers or even lower cost materials)
- Increase barrier properties: clays and graphene both hold promise here, especially with flammability
    - Flammability being driven by trying to eliminate brominated compounds
- Increase thermal conductivity: carbon nanotubes and graphene (doesn’t work very well; won’t discuss why explicitly in this talk)
- Increase electrical conductivity: carbon nanotubes (successful) and graphene (competition: carbon black)

NANOCLAYS

What is dispersion of nanoclays?

- Clays are formed of stacked ~1 nm sheets (platelets)
- Need to cause the sheets to exfoliate (become individual sheets). Disperse clay in water, add cationic surfactant (opposite charge of clay) to increase distance between platelets and weaken interplatelet force, dry, then use high shear force to disperse.

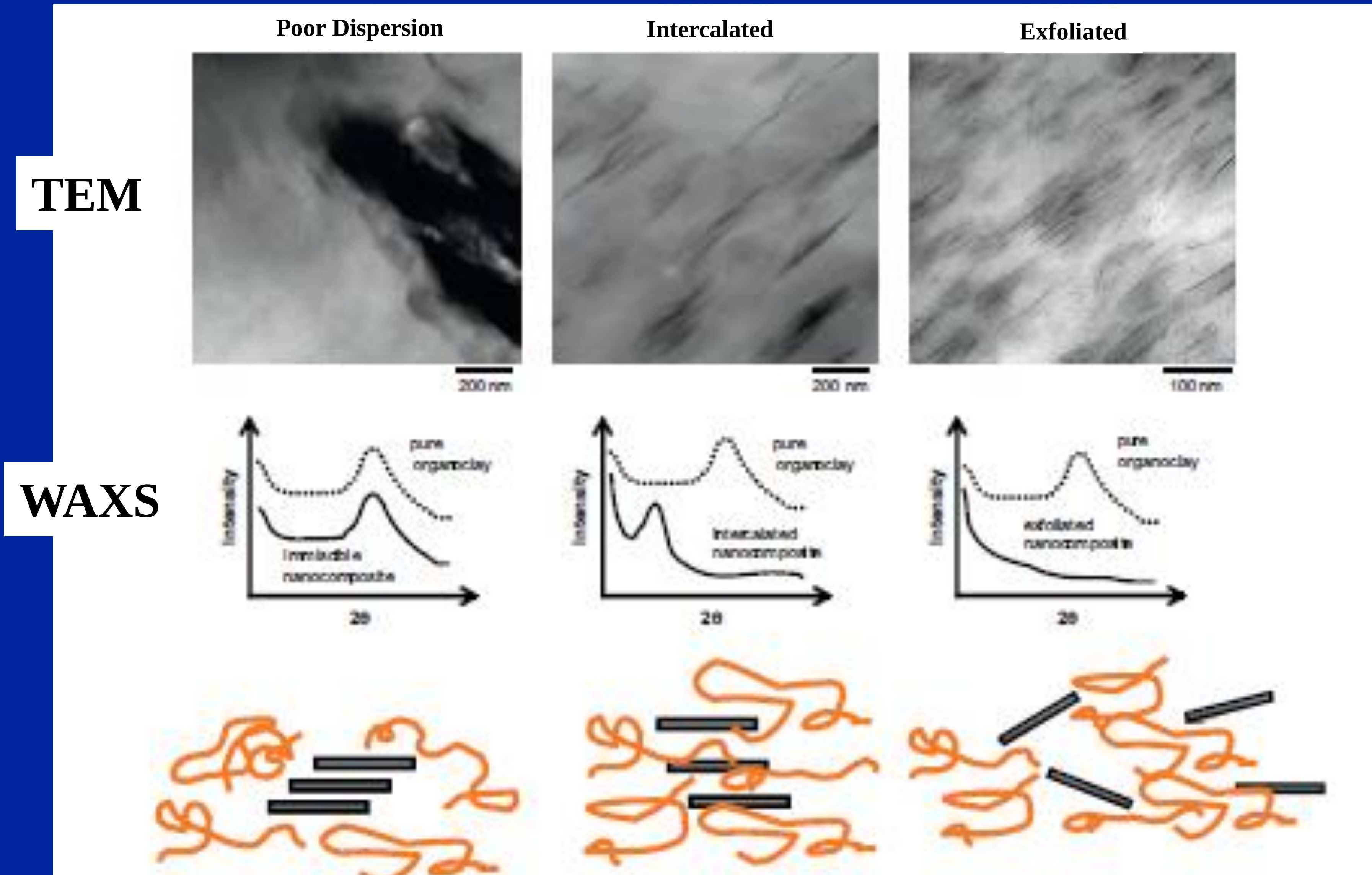


Paul et.al. Polymer, 49 (2008) 3187

How to disperse nanoclays?

- Key is choice of surfactant
  - For more polar polymers (e.g. nylon, polycarbonate), use single-tailed surfactants; for less polar polymers (polyethylene, polypropylene), use double-tailed surfactants
- Use twin-screw extrusion to disperse nanoclays
  - Peeling of layers, rather than diffusion of polymers in between layers, seems to be the mechanism
  - Much better dispersion of nanoclays into polar polymers (high surface energy) then non-polar (low surface energy) polymers
  - Want high shear rate, high viscosity
- Dispersion into single layers still limits clays, especially in polyolefins

How measure dispersion of nanoclays?



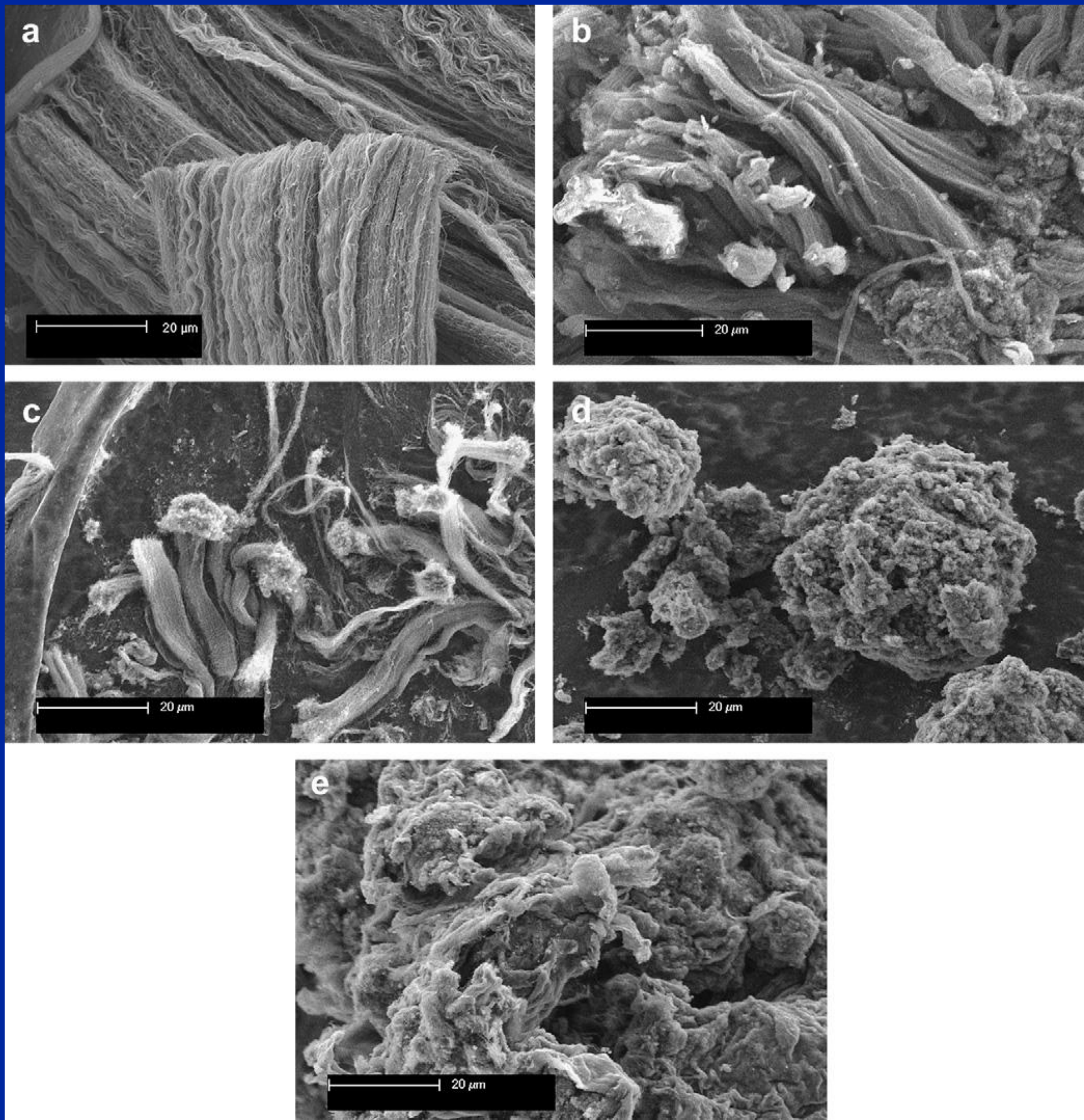
Notes about measuring dispersion

- Lack of peak in diffraction pattern does NOT mean full exfoliation
- To get excellent TEM pictures requires orienting the sample so that edge on images are obtained

CARBON NANOTUBES

What is dispersion of nanotubes

- Tubes are usually found as a powder.
- Think of a ball of yarn
- Need to untangle tubes from this ball of yarn
- Lower density powders disperse better*



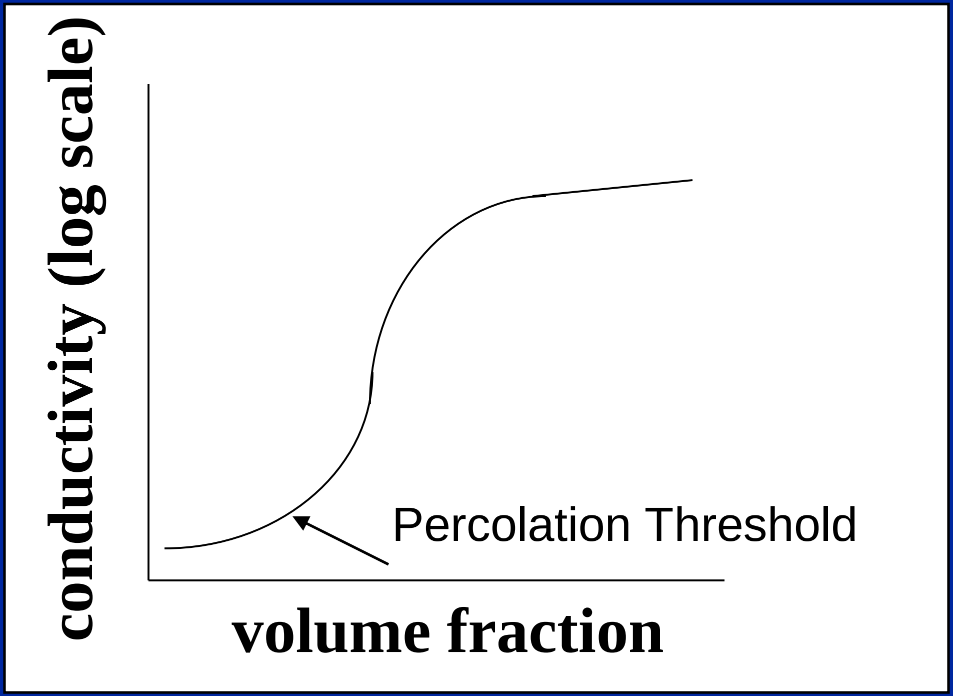
*Morcom et.al. Polymer, 51 (2010) 3540

How to disperse nanotubes in polymers?

- MWCNT can be dispersed by melt-mixing (twin-screw extrusion)
  - Synthesis key, i.e synthesizing well-dispersable MWCNTs. By synthesis, I mean what you buy from the manufacturer (i.e. can include purification steps)
  - Want high shear, high rotation rate. Will break tubes. However, work in our lab has shown for long tubes, if you are going for electrical conductivity there is an optimum in terms of minimizing amount needed to conduct and maximizing conductivity
- Infiltration/Infusion approach (mats, fibers) for liquid monomers
  - Use nanotubes dispersed in a liquid and spray the nanotubes into mats or draw the nanotubes into fibers. Add liquid monomer and polymerize.

How measure dispersion?

- Percolation methods (usually electrical)
  - At what volume fraction is the percolation threshold? Lower=better dispersion
  - Who says that dispersion at higher volume fractions is the same as at lower volume fractions?



- Microscopy-very labor intensive
  - SEM or TEM typical composite approach; also unique approach
  - Optical Microscopy- Quantitative for macrodispersion

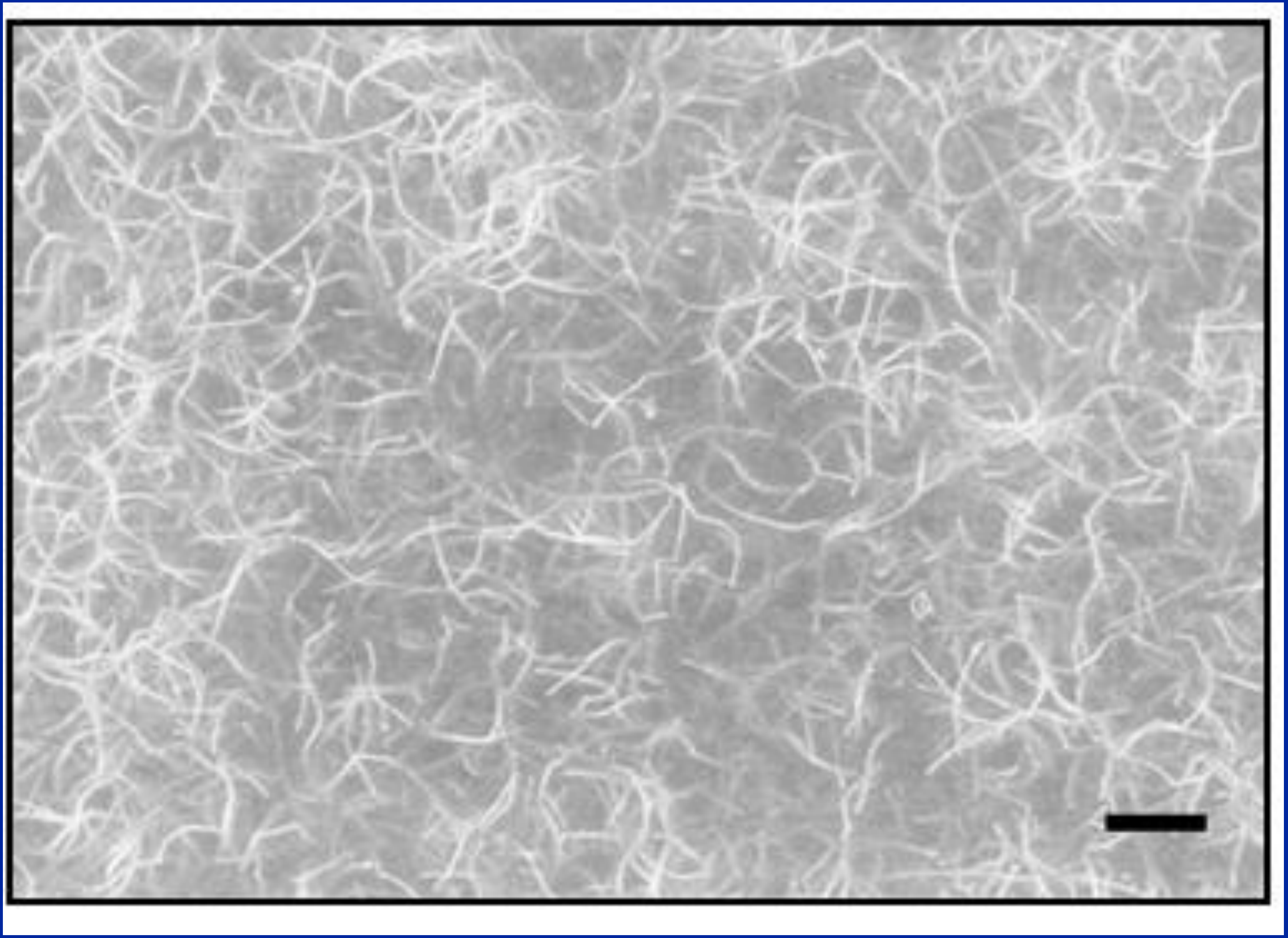
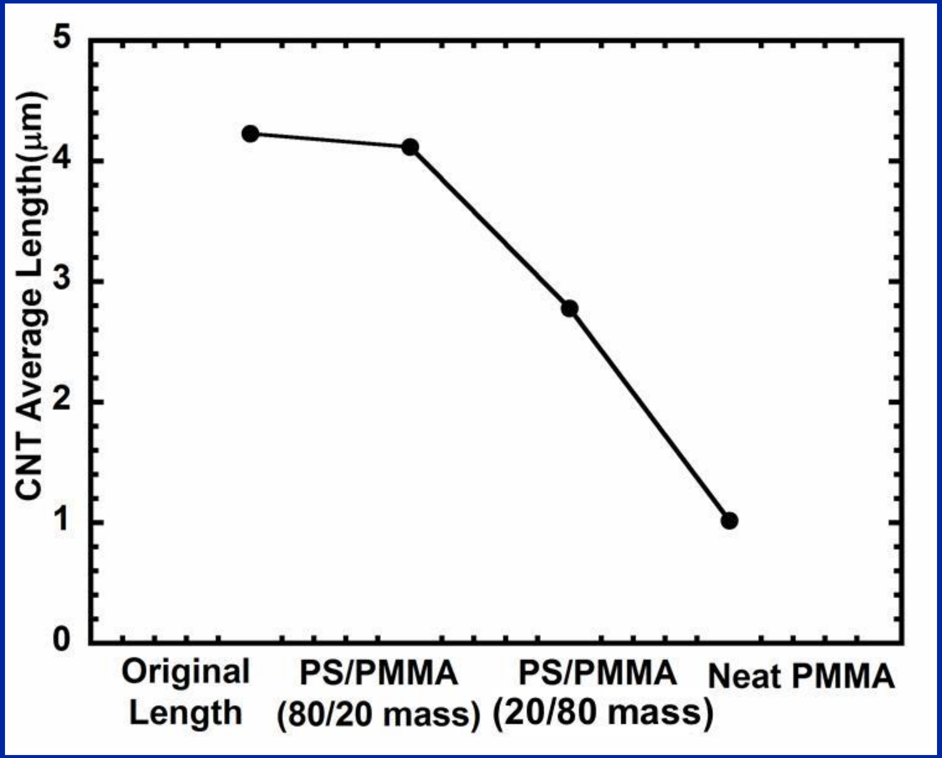
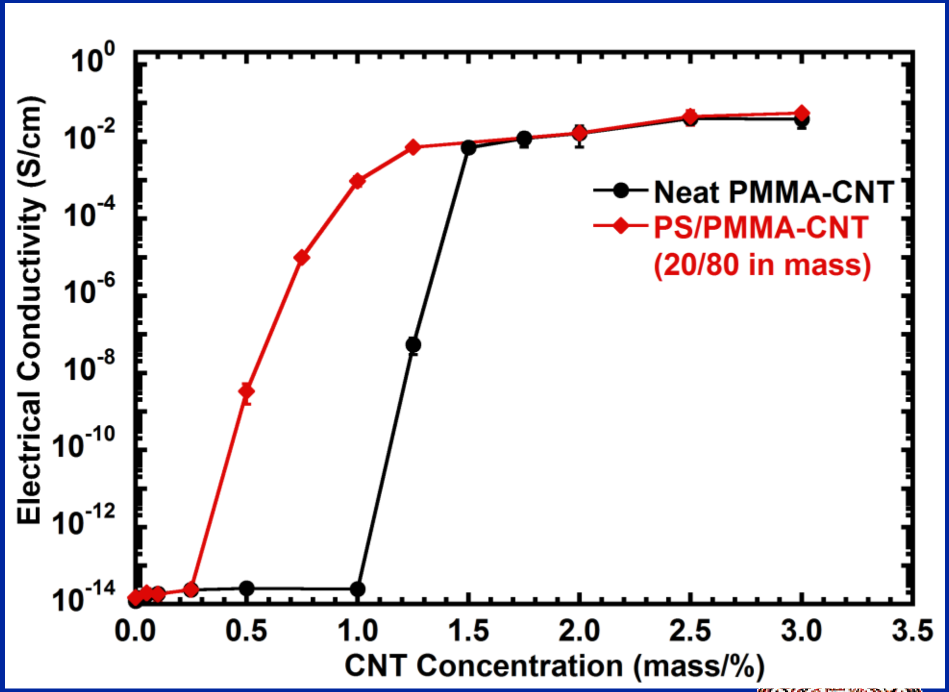


Image: Lillehei et.al. Nanotechnology, 20, 325708 (2009)
Scale bar is 2 microns

Effect of Blend Ratio on Nanotube Length/Conductivity





GRAPHENE

What is dispersion of graphene?

- At this point, most methods that could produce graphene in ton quantities start with graphite.
- In terms of mixing with polymers: I think it is entirely appropriate to think of graphite as clays that are harder to disperse
  - Can't just add graphene to a solution and add a molecule to separate the layers
- Oxidize with an acid partially that doesn't make highly-substituted graphene oxide and then heat to expand layers.
  - Get a few sheet thick graphene
- Graphene oxide
  - Treat with strong acid and potassium permanganate (Hummers method)
  - Exfoliation occurs
  - Can reduce back to graphene and maintain exfoliation

How to disperse graphene in polymers?

- Treated graphene or graphene oxide can be dispersed by melt-mixing (twin-screw extrusion)
  - Lose conductivity with graphene oxide
- Latex approaches might hold some promise
  - Graphene oxide can be dispersed reasonably in water; graphene can be dispersed with stabilization agent
  - Add this to latex paint
  - Why not nanotubes or clays
    - vs. clays; the surfactant in the clay can deadsorb and combine with the surfactant in the latex
    - vs. nanotubes: nanotubes will affect the rheology of the latex to a much higher extent than graphene

How measure dispersion?

- Percolation methods (usually electrical)
- Wide angle x-ray diffraction (WAXS)
- TEM
  - Graphene is harder to get good images then clays because contrast is less

- Nanoclays
  - Cost they are now is the cost that they will be in the future
  - Marginally useful in polymers, mature market
  - Dispersion is solved, but solution is expensive.
- Nanotubes
  - Fundamentally, the cost of multiwalled nanotubes could be as cheap as polyethylene. Monomers are cheap, it is all about yield (pounds of product/pounds of catalyst).
  - Still a growing market
  - Dispersion is solved, but knowledge is not complete
- Graphene
  - Unclear whether graphite can be made into graphene cost-effectively ever.
  - Synthesize graphene in large scales cost effectively?
  - Market hasn’t really developed; will it ever?
  - Cost-effective dispersion is key to graphene in polymers.